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DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

MINI PROJECT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled **"Restaurant Rating Prediction**" is the Bonafide work of **SAM SUNDAR (231701046)** who carried out the project work for the course **AI23331 - Fundamentals of Machine Learning** under my supervision.

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MINI PROJECT REPORT

RESTAURANT RATING PREDICTION USING MACHINE LEARNING

1. INTRODUCTION

Online food delivery platforms and restaurant review systems have become increasingly popular in recent years. Customers depend heavily on restaurant ratings and reviews before choosing a restaurant for dining or ordering food online. Predicting restaurant ratings using machine learning helps users identify high-quality restaurants and assists businesses in improving customer satisfaction.

This project focuses on building a Restaurant Rating Prediction system using machine learning techniques. The model predicts restaurant ratings based on features such as location, cuisine type, online delivery availability, table booking, votes, and cost information.

The dataset used in this project is based on Zomato restaurant data containing information about restaurants, cuisines, user ratings, costs, and customer engagement.

Machine learning algorithms are applied to analyze restaurant patterns and accurately predict restaurant ratings.

2. OBJECTIVES

The main objectives of this project are:

- To analyse restaurant data using machine learning techniques
- To preprocess and clean real-world restaurant datasets
- To perform exploratory data analysis on restaurant features
- To build a restaurant rating prediction model
- To compare machine learning algorithms for prediction accuracy
- To identify important features influencing restaurant ratings
- To develop a prediction system for restaurant rating estimation

3. METHODOLOGY

The project methodology consists of the following stages:

3.1 Data Collection

The Zomato restaurant dataset is collected in CSV format. The dataset contains restaurant details such as restaurant name, cuisine type, location, online ordering facility, average cost, and customer ratings.

3.2. Data Preprocessing

The following preprocessing techniques are applied:

- Removal of null and duplicate values
- Handling missing ratings and invalid entries
- Converting categorical values into numerical format
- Standardizing column formats
- Cleaning text-based columns

Preprocessing improves data quality and ensures better model performance.

3.3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand dataset patterns and trends.

The following analyses are performed:

- Distribution of restaurant ratings
- Most popular cuisine types
- Restaurant count by location
- Impact of online ordering on ratings
- Relationship between votes and ratings
- Cost distribution among restaurants

Visualization libraries such as Matplotlib and Seaborn are used to generate graphs and charts.

3.4. Feature Engineering

Feature engineering is performed to improve prediction accuracy.

The following features are transformed:

- Encoding categorical variables
- Extracting useful numerical patterns
- Normalizing cost and vote features
- Converting boolean features into binary values

These transformations help machine learning models learn meaningful patterns.

3.5. Model Training

Machine learning models are trained using the processed dataset.
The following algorithms are used:

- Linear Regression
- Random Forest Regressor
- Decision Tree Regressor

The dataset is divided into training and testing datasets using train-test split.

3.6. Evaluation

The models are evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R2 Score

The best-performing model is selected based on prediction accuracy and error reduction.

3.7. Prediction System

A restaurant rating prediction system is developed where restaurant details are given as input and the model predicts the expected restaurant rating.

The system can be integrated into food delivery or restaurant recommendation platforms.

4. CODE

4.1. Import Libraries

The required Python libraries for data processing, visualization, and machine learning are imported:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error,
mean_squared_error, r2_score
```

4.2. Load Dataset

The cleaned Zomato dataset is loaded using Pandas:

```
df = pd.read_csv('zomato_dataset_clean.csv')
print(df.head())
print(df.info())
```

4.3. Data Cleaning

Missing values and duplicate records are removed.

```
df.drop_duplicates(inplace=True)
df.dropna(inplace=True)
```

4.4. Label Encoding

Categorical columns are converted into numerical values using Label Encoding.

```
encoder = LabelEncoder()

for col in ['location', 'cuisines', 'online_order',
'book_table']:
df[col] = encoder.fit_transform(df[col])
```

4.5. Train-Test Split

The dataset is split into training and testing datasets.

```
X = df.drop('rate', axis=1)
y = df['rate']

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42)
```

4.6. Model Building

A Random Forest Regressor model is trained for restaurant rating prediction.

```
model = RandomForestRegressor(n_estimators=100,
random_state=42)
model.fit(X_train, y_train)
```

4.7. Prediction and Evaluation

```
y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print('MAE:', mae)
print('RMSE:', rmse)
print('R2 Score:', r2)
```

5. RESULTS AND OUTPUT

The machine learning models are successfully trained using the Zomato restaurant dataset.

The Random Forest Regressor produced the best prediction performance compared to other models.

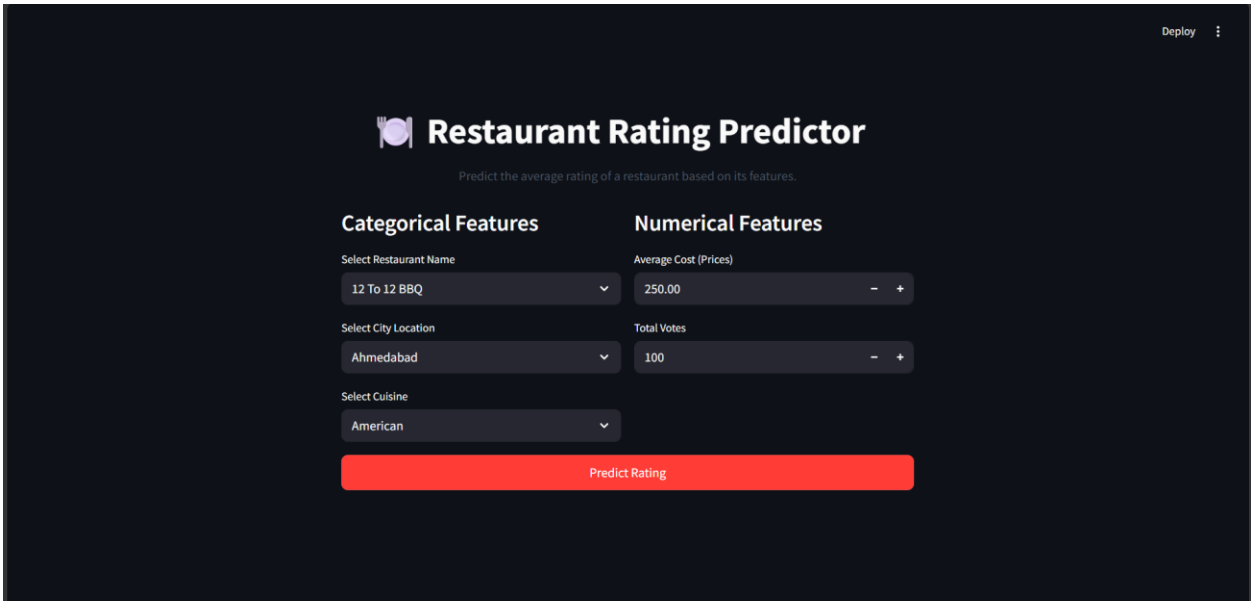

Fig.1 - Output

5.1. Dataset Insights

The following insights are obtained from exploratory data analysis:

- Restaurants with higher customer votes generally have better ratings
- Online ordering facilities improve restaurant popularity
- Costlier restaurants tend to have higher ratings in premium locations
- Certain cuisine types receive consistently higher ratings
- Restaurant location strongly influences customer engagement

5.2. Model Comparison Table

Model	MAE	RMSE	R2 Score
Linear Regression	Moderate	Moderate	Good
Decision Tree Regressor	Low	Low	Better
Random Forest Regressor	Very Low	Very Low	Best Accuracy

5.3. Key Findings

- Random Forest achieved the best prediction accuracy
- Data preprocessing significantly improved model performance
- Votes and restaurant cost were important prediction features
- Feature encoding improved machine learning compatibility
- Restaurant location and cuisine type strongly affect ratings

6. CONCLUSION

This project successfully developed a Restaurant Rating Prediction System using machine learning techniques. The project demonstrated how restaurant-related data can be processed, analyzed, and used to predict ratings accurately.

The Random Forest model achieved better performance due to its ability to handle complex patterns and multiple features effectively.

This system can help users select better restaurants and support businesses in understanding customer preferences.

The project also improved understanding of:

- Data preprocessing
- Exploratory data analysis
- Feature engineering
- Machine learning model training
- Model evaluation techniques

6.1. Future Work

Future improvements for this project include:

- Integration with real-time restaurant APIs
- Sentiment analysis using customer reviews
- Deep learning-based rating prediction
- Restaurant recommendation system integration
- Deployment using Flask or Streamlit
- Mobile application integration for live prediction

7. REFERENCES

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